Lesson plan · for the teacher

Whole numbers and yes-or-no choices

Optimization · Level 2 · Lesson 3

Time	Materials
40 min	Board and chalk; A table on paper for each group: buses 0–3 across, minibuses 0–5 down; The trip numbers on the board: 64 people, seats and prices

Learning goals

- Say which decisions must be whole numbers or yes-or-no.
- Show how rounding a fraction can break a limit or cost more.
- Find the cheapest whole-number plan for our trip.

Activity: Buses for the trip

1. Warm-up (5 min): Write the trip numbers on the board and work out 64 ÷ 30 = 2.13 together. Ask: can we hire 2.13 buses? What should we do?
2. Teach (8 min): Explain that buses come whole and that some choices are yes or no. Show what rounding does: 2 buses leave 4 people behind; 3 buses cost 1,650.
3. Table (15 min): Groups fill a table, buses 0–3 across and minibuses 0–5 down, with the seats and cost in each cell, and cross out cells with fewer than 64 seats.
4. Compare (7 min): Groups circle the cheapest plan that seats everyone (2 buses and 1 minibus, 1,400) and compare it with both roundings. Which was closer?
5. Fair? (5 min): Ask who should ride the minibus, and whether every class gets to sit together. Is the cheapest plan the fairest?

Prepare one blank table per group. Use your own trip's numbers if you have them: the rounding problem appears with almost any numbers.

Discussion

- Why was the best whole-number plan not near either rounding?
- Which decisions at school are yes or no?
- Who should decide the points when we pack supplies for another school?

Exit ticket

- Give one decision that must be a whole number, and say what rounding could get wrong.

Try it at school tomorrow

Plan the trip buses with your pupils: 64 people, minibuses of 14 seats for 300 coins and buses of 30 seats for 550. Groups list whole-number plans in a table, find the cheapest that seats everyone, and compare it with rounding 2.13 buses up or down.

Practice questions · for you

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1 **The village school says the water filter must go on the truck, whatever its points. In the yes-or-no model, what should change?**

- ◯ A new limit: take-filter = 1
- ◯ The goal: the filter's points count double
- ◯ Nothing: the solver will take it anyway

2 **Rounding the linear program's answer up always gives the cheapest whole-number plan.**

- ◯ True
- ◯ False

3 **What does this program print, and what is wrong with the plan?**

```
people = 64
seats = 30
buses = people / seats
print(round(buses))
```

- ◯ 3, and everyone has a seat
- ◯ 2.13, so hire part of a bus
- ◯ 2, and four people have no seat

4 **Complete the truck model: one yes-or-no variable per load, the weight limit (in hundreds of kilos) and the goal.**

```
from ortools.sat.python import cp_model
W = [6, 5, 5, 3, 2, 1]
P = [42, 33, 32, 15, 8, 3]
R = range(6)
m = cp_model.CpModel()
t = [m. _____ ("") for i in R]
m.add(sum(W[i]*t[i] for i in R) _____ 10)
m. _____ (sum(P[i]*t[i] for i in R))
```

Word bank: `>=` `new_constant` `new_bool_var` `<=` `maximize` `minimize`

5 **Match each decision to the kind of variable it needs.**

How many buses to hire	• • • •	Any number, fractions too
Whether to take the solar panels	• • • •	A whole number: 0, 1, 2, …
Kilograms of rice to buy for the kitchen	• • • •	Yes or no: 0 or 1

✎ **Which decisions at your school must be whole numbers or yes-or-no? Who sets the "points" for them, and who should?**

Check your answers

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1 **A new limit: take-filter = 1**

A rule that must always hold is a constraint: take_filter = 1.

2 **False**

Rounding 2.13 buses up gives 3 buses for 1,650 coins, but 2 buses and 1 minibus seat everyone for 1,400.

3 **2, and four people have no seat**

64 / 30 = 2.13, and round() gives 2. Two buses seat 60, so 4 of the 64 stay behind: rounding to the nearest whole number broke the seating limit.

4

```
from ortools.sat.python import cp_model
W = [6, 5, 5, 3, 2, 1]
P = [42, 33, 32, 15, 8, 3]
R = range(6)
m = cp_model.CpModel()
t = [m.new_bool_var("") for i in R]
m.add(sum(W[i]*t[i] for i in R) <= 10)
m.maximize(sum(P[i]*t[i] for i in R))
```

new_bool_var makes a variable that is 0 or 1, take it or leave it.

5

How many buses to hire — A whole number: 0, 1, 2, …

Whether to take the solar panels — Yes or no: 0 or 1

Kilograms of rice to buy for the kitchen — Any number, fractions too

Buses come whole; a load is taken or not; rice can be weighed in any amount, so a fraction is fine.